Quantum Computing

UNIT - I: Mathematical Foundations for Quantum Systems

Lecture Notes (with Figures and Key Equations)

Unit Overview

Quantum computing is built on linear algebra over complex vector spaces. In this unit, we develop the mathematical language required to represent quantum states and model quantum gates as linear (unitary) operators. The focus is on concepts that directly appear in quantum algorithms: basis representations, inner products, matrix operations, and eigen-decomposition.

Learning outcomes (after completing Unit - I, you should be able to):

1) Represent quantum states as vectors in a complex Hilbert space.
2) Compute inner products, norms, and probabilities of measurement outcomes.
3) Perform matrix operations (transpose, conjugate, adjoint) and interpret matrices as transformations.
4) Find eigenvalues/eigenvectors, diagonalize important matrices, and connect them to quantum observables.
5) Apply common quantum gates (X, Z, H) as unitary matrices acting on state vectors.

1. Vector spaces, basis, and dimension

A vector space V over a field (here, usually the complex numbers C) is a set of objects called vectors that can be added together and multiplied by scalars, satisfying standard axioms (closure, associativity, distributivity, existence of identity and inverses). In quantum computing, the state space of an n-qubit system is a complex vector space of dimension 2^n.

Key definitions:

- Linear combination: a1 v1 + a2 v2 + ... + ak vk, where ai are scalars and vi are vectors.
- Span: the set of all linear combinations of a given set of vectors.
- Basis: a set of vectors that is linearly independent and spans the space.
- Dimension: the number of vectors in any basis of the space.

Worked idea:

If {e1, e2} is a basis for a 2D space, every vector v can be written uniquely as v = x e1 + y e2. The numbers (x, y) are the coordinates of v in that basis.

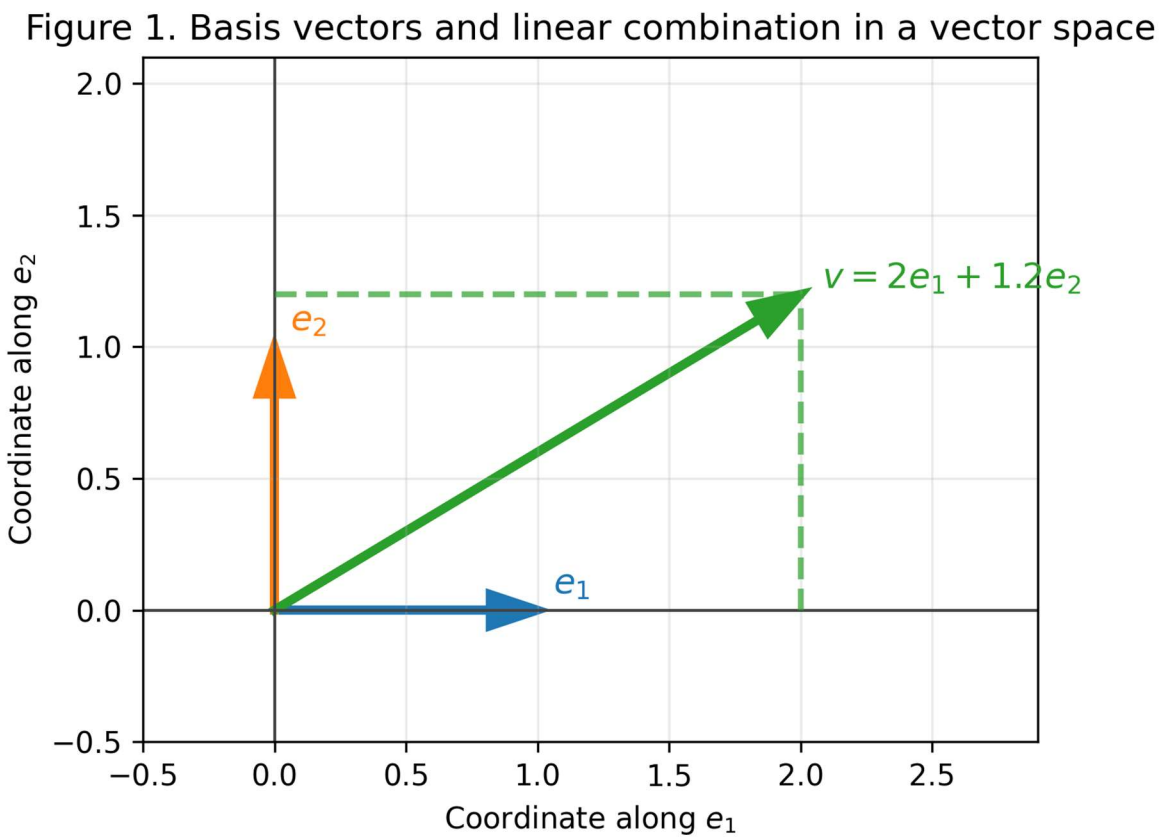


Figure 1 shows how a vector is expressed as a linear combination of basis vectors.

Important note for quantum systems:

The computational basis for one qubit is { |0>, |1> }. For n qubits, the computational basis has 2^n vectors such as |00...0>, |00...1>, ..., |11...1>.

2. Inner product, orthogonality, and Hilbert spaces

An inner product provides a way to measure angles and lengths. For complex vector spaces, the inner product must include complex conjugation to behave well. A complete inner-product space is called a Hilbert space. Quantum states live in Hilbert spaces.

Complex inner product (finite-dimensional):

$$\langle \phi | \psi \rangle = \sum_i \phi_i^* \psi_i$$

Eq. (1) Complex inner product (bra-ket notation).

From the inner product we define:

- Norm (length): ||psi|| = sqrt(<psi|psi>).
- Orthogonality: vectors u and v are orthogonal if <u|v> = 0.
- Normalization: a state vector |psi> must satisfy <psi|psi> = 1.

$$\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$$

Eq. (2) Norm derived from the inner product.

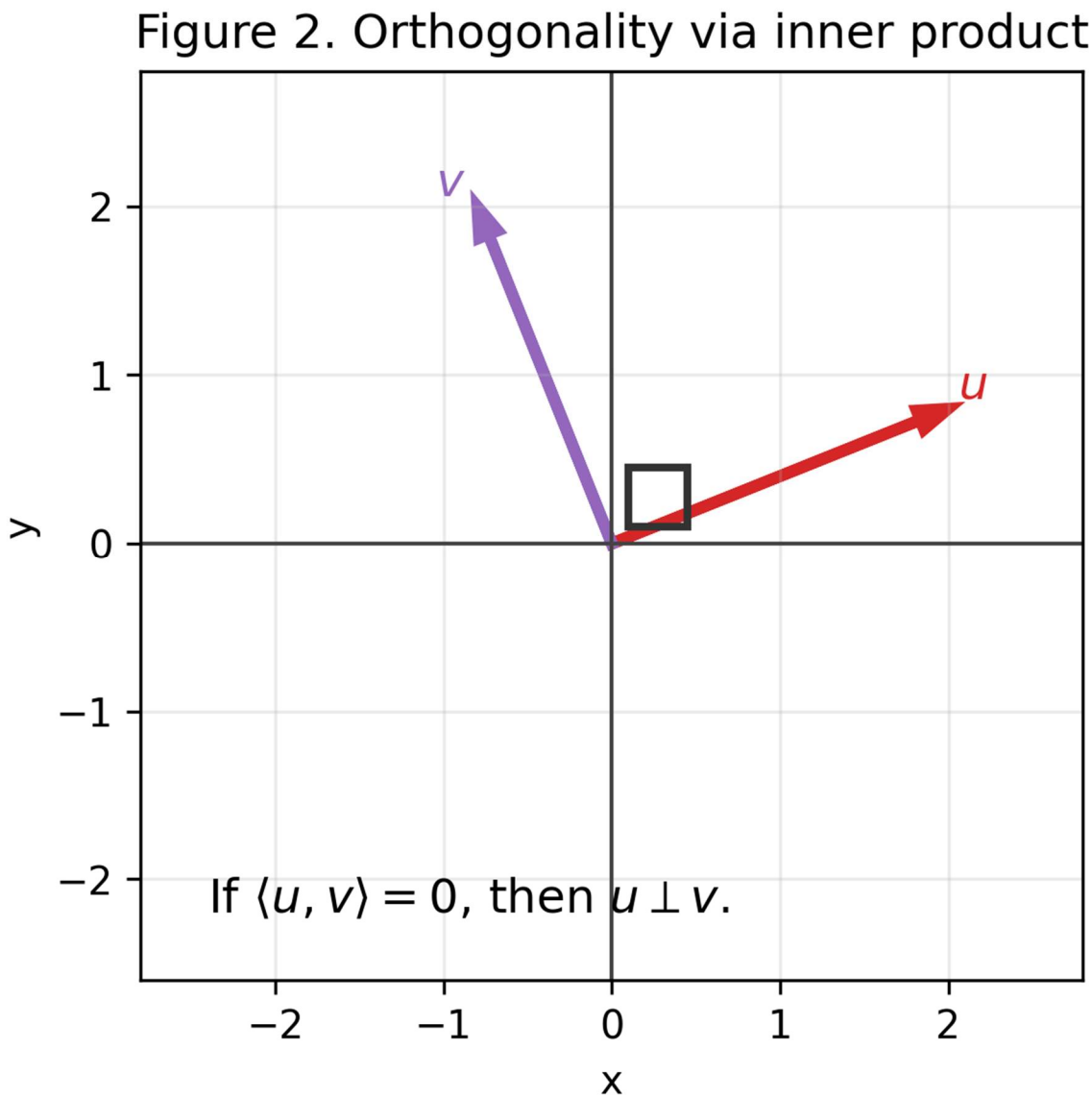


Figure 2 illustrates orthogonality: <u|v> = 0 implies a right angle.

Hilbert space (intuition):

Think of a Hilbert space as a vector space with a notion of distance and angle, where limits of sequences (completeness) make sense. In practical quantum computing, we often work with finite-dimensional Hilbert spaces such as C^2 (one qubit) or C^{2^n} (n qubits).

Figure 3. Qubit as a vector in a 2D Hilbert space (computational basis)

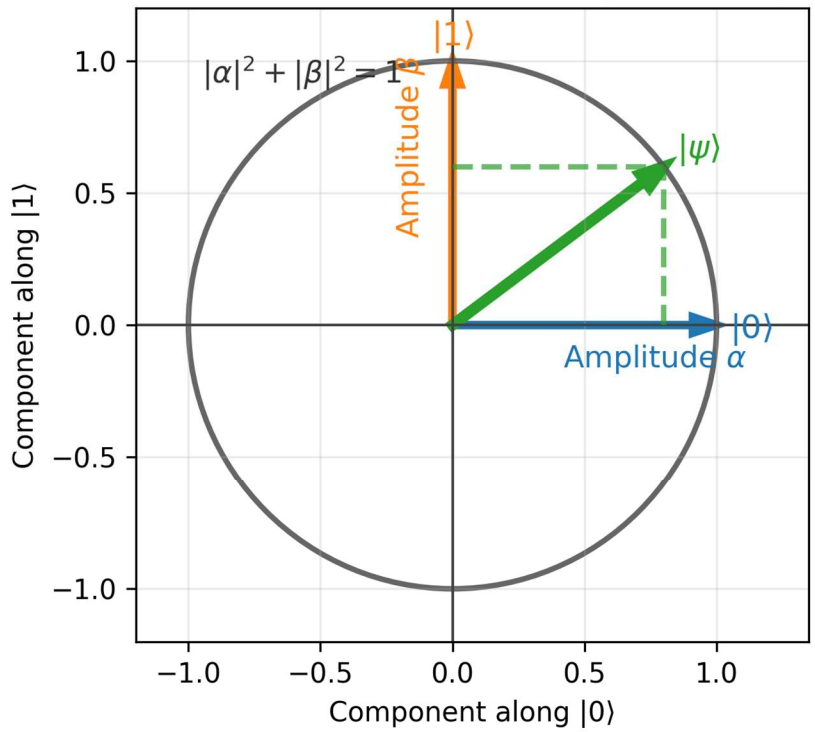


Figure 3: a qubit is a normalized vector in a 2D complex Hilbert space.

3. Complex matrices and linear transformations

A linear transformation T: V -> V can be represented by a matrix A once we fix a basis. For complex matrices we use the conjugate transpose (also called the adjoint or dagger): A^† = (A^*)^T. This operation is central in quantum computing because quantum gates must be unitary.

Matrix operations you must know:

- Transpose: A^T (swap rows and columns).
- Complex conjugate: A^* (take complex conjugate of every entry).
- Conjugate transpose (adjoint): A^† = (A^*)^T.
- Identity matrix: I (acts like 1 for matrix multiplication).

Unitarity (quantum gate condition):

$$U^\dagger U = I \quad \text{(unitary)}$$

Eq. (3) A gate U is unitary if U^† U = I (preserves norms/probabilities).

Interpretation: applying a unitary matrix U to a state |psi> produces a new state |psi' > = U|psi>. Because U preserves inner products, total probability remains 1.

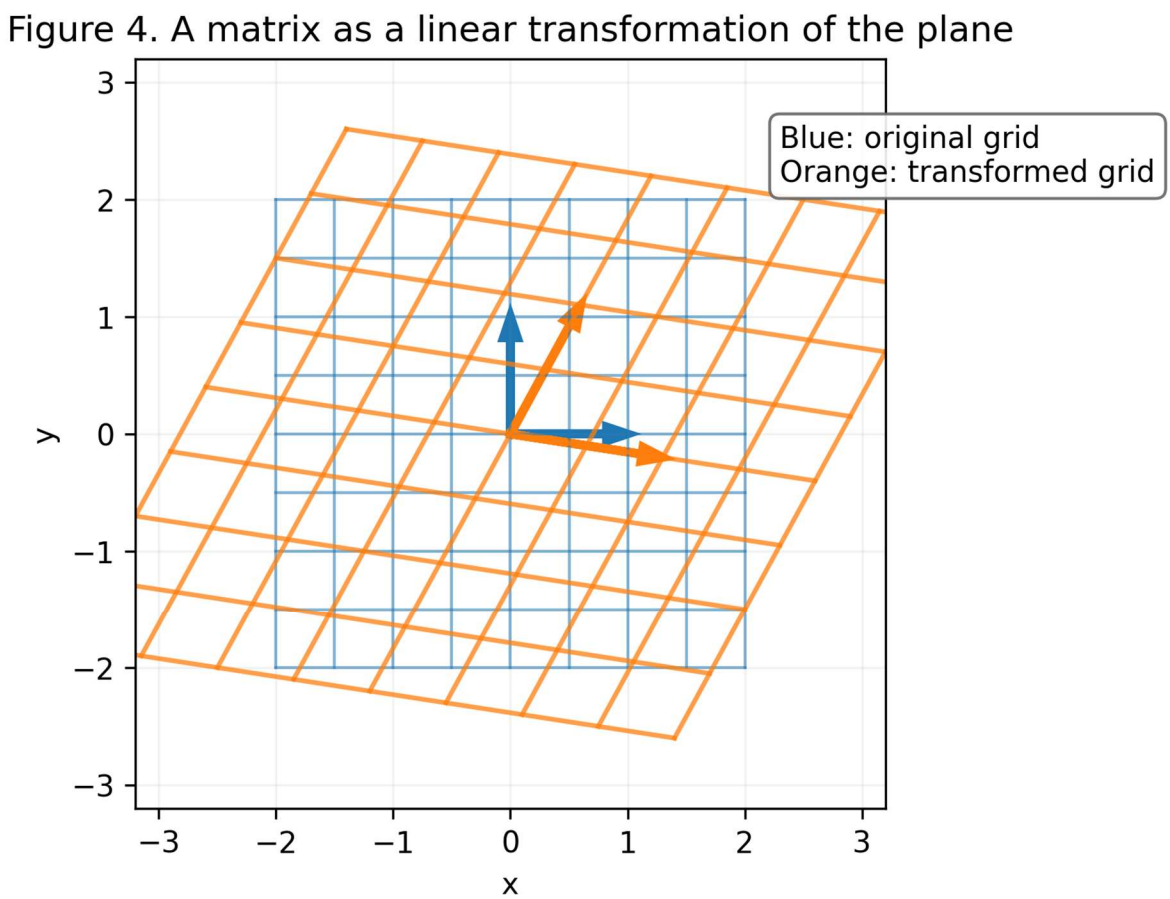


Figure 4: a matrix transforms every vector and reshapes the coordinate grid.

4. Eigenvalues, eigenvectors, and diagonalization

Eigenvectors are special directions that are preserved by a linear transformation. If Av = λv, then v is an eigenvector of A with eigenvalue λ. Eigen-decomposition helps us understand powers of

matrices and is directly connected to quantum measurement: many observables are represented by Hermitian matrices whose eigenvalues are the possible measurement outcomes.

Finding eigenvalues (concept):

- Solve det(A - λI) = 0 to obtain eigenvalues λ.
- For each eigenvalue λ, solve (A - λI)v = 0 to obtain eigenvectors v.

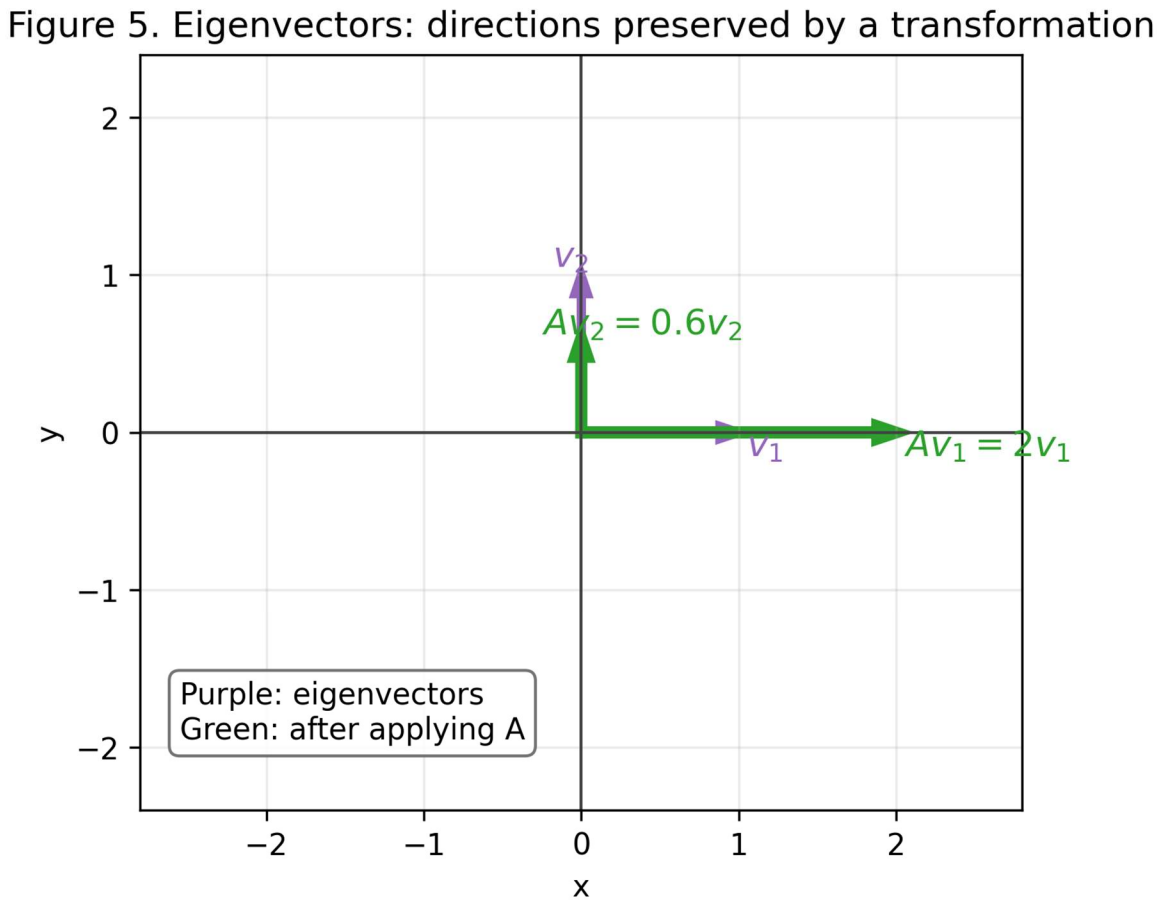


Figure 5: eigenvectors keep their direction; eigenvalues tell the scaling.

Diagonalization (why it matters):

If A can be diagonalized, computations become simple because diagonal matrices are easy to power and analyze. For a diagonalizable matrix A, we can write A = VΛV^{-1}, where Λ is diagonal (eigenvalues) and V contains eigenvectors.

$$A = U \Lambda U^{\dagger} \quad (A \text{ Hermitian})$$

Eq. (4) Spectral form for a Hermitian matrix A (important for observables).

Example (Pauli-Z):

The Pauli-Z matrix is Z = [[1, 0], [0, -1]]. Its eigenvalues are +1 and -1. The corresponding eigenvectors are |0> for +1 and |1> for -1. This matches the idea that measuring along the computational basis yields two possible outcomes.

5. Applications to quantum state representation and gate operations

A single qubit state is a normalized vector in C^2. Using the computational basis, any pure state can be written as a superposition of |0> and |1> with complex amplitudes.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Eq. (5) Superposition state of one qubit.

Normalization and measurement probabilities:

- Normalization: <psi|psi> = 1 implies |α|^2 + |β|^2 = 1.
- Measurement in the computational basis returns 0 with probability |α|^2 and 1 with probability |β|^2.

$$P(0) = |\alpha|^2,\ P(1) = |\beta|^2$$

Eq. (6) Born rule for measurement probabilities (computational basis).

Quantum gates as matrices:

Quantum gates are linear operators represented by unitary matrices. Common single-qubit gates include:

Gate	Matrix	Effect (intuition)
X (NOT)	[[0, 1], [1, 0]]	Flips |0> <-> |1>
Z	[[1, 0], [0, -1]]	Adds a phase: |1> -> -|1>

H (Hadamard)	(1/√2)*[[1, 1], [1, -1]]	Creates/undoes superposition

Geometric picture (Bloch sphere):

Figure 6. Bloch sphere view of a single-qubit state

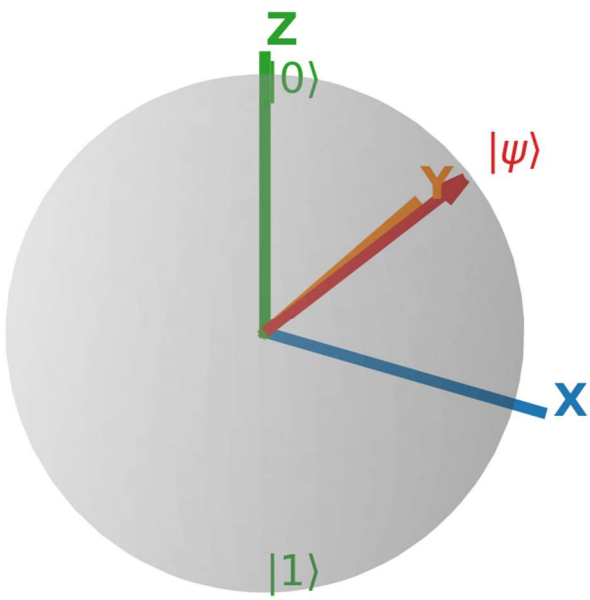
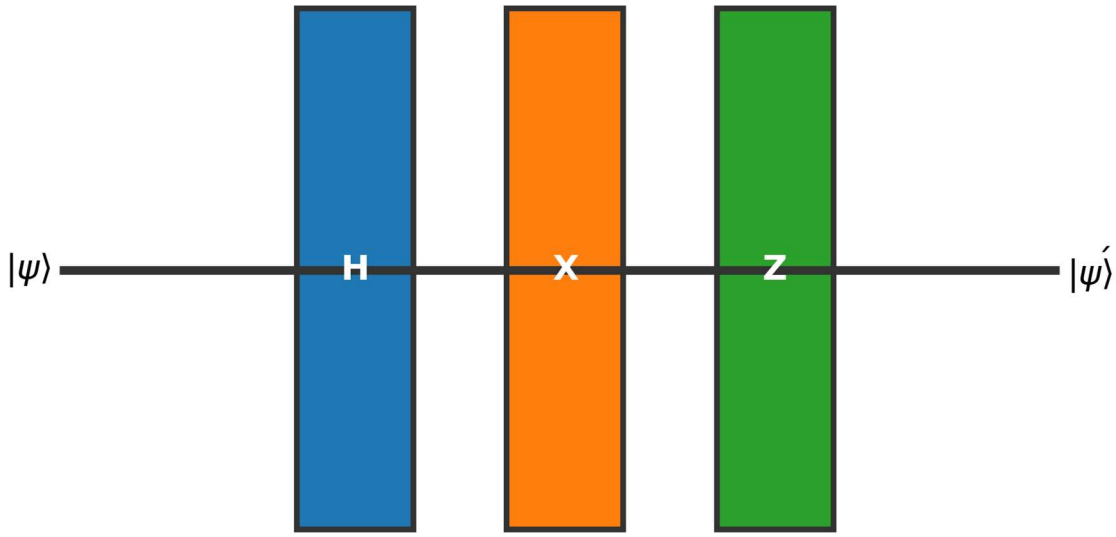


Figure 6 provides a geometric view of one-qubit pure states on the Bloch sphere.

Circuit-to-math correspondence:

Figure 7. Quantum circuit as matrix multiplication on state vectors



Equivalent math: $|\psi'\rangle = Z X H |\psi\rangle$

Figure 7: applying gates in a circuit equals multiplying matrices (right to left).

Worked example 1 (Hadamard on |0>):

Represent |0> as the column vector [1, 0]^T. Then apply H:

H|0> = (1/√2)[[1, 1],[1, -1]] [1, 0]^T = (1/√2)[1, 1]^T = (|0> + |1>)/√2.

Worked example 2 (unitarity preserves probabilities):

Let |psi> be any normalized state. Because U is unitary, <psi|psi> = 1 implies <psi|U^†U|psi> = 1, so the transformed state U|psi> is also normalized. This is why quantum gates do not change total probability.

Summary Cheat Sheet

Core formulas to remember:

- Superposition: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
- Inner product: $\langle\phi|\psi\rangle = \sum_i \phi_i^* \psi_i$
- Norm: $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$
- Unitary gate: $U^\dagger U = I$
- Eigen equation: $Av = \lambda v$
- Hermitian diagonalization: $A = U\Lambda U^\dagger$
- Measurement probs: $P(i) = |\text{amplitude}_i|^2$

Practice Questions (Quick Check)

1. In C^2, show that the vectors $|0\rangle = [1,0]^T$ and $|1\rangle = [0,1]^T$ form a basis.
2. Compute $\langle u|v\rangle$ for $u = [1, i]^T$ and $v = [i, 1]^T$. Are they orthogonal?
3. Verify that the Pauli-X matrix is unitary.
4. Find eigenvalues and eigenvectors of $Z = [[1,0],[0,-1]]$.
5. If $|\psi\rangle = (1/\sqrt{3})|0\rangle + (\sqrt{2}/\sqrt{3})|1\rangle$, compute $P(0)$ and $P(1)$.

Answer hints (brief):

1) They are linearly independent and span C^2.
2) Use $\langle u|v\rangle = u^\dagger v$ (remember conjugate).
3) Check $X^\dagger X = I$ (for X, $X^\dagger = X$).
4) Eigenvalues are +1 and -1 with eigenvectors $|0\rangle$ and $|1\rangle$.
5) Probabilities are squares of magnitudes: $P(0)=1/3$, $P(1)=2/3$.

End of Unit - I Notes